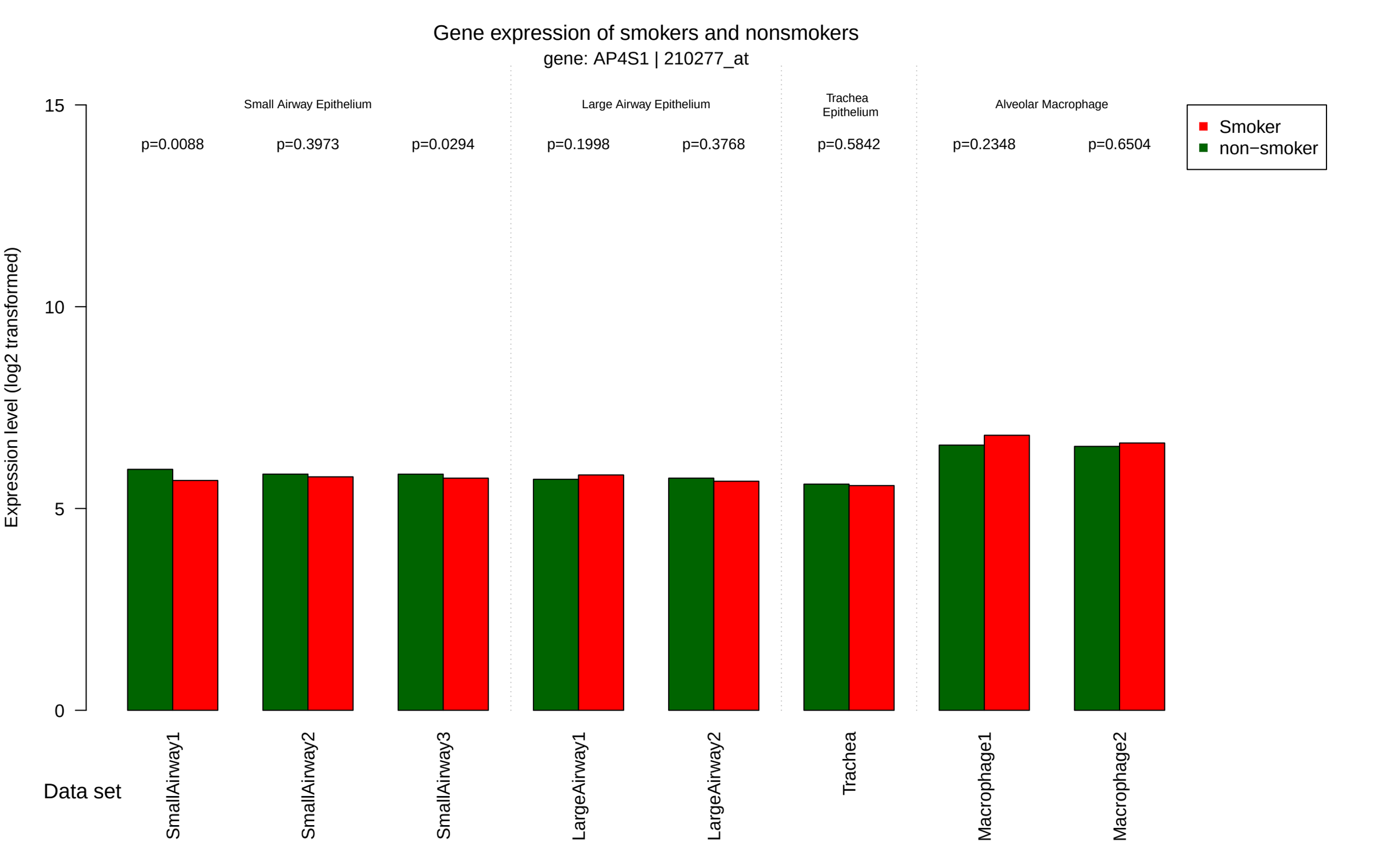
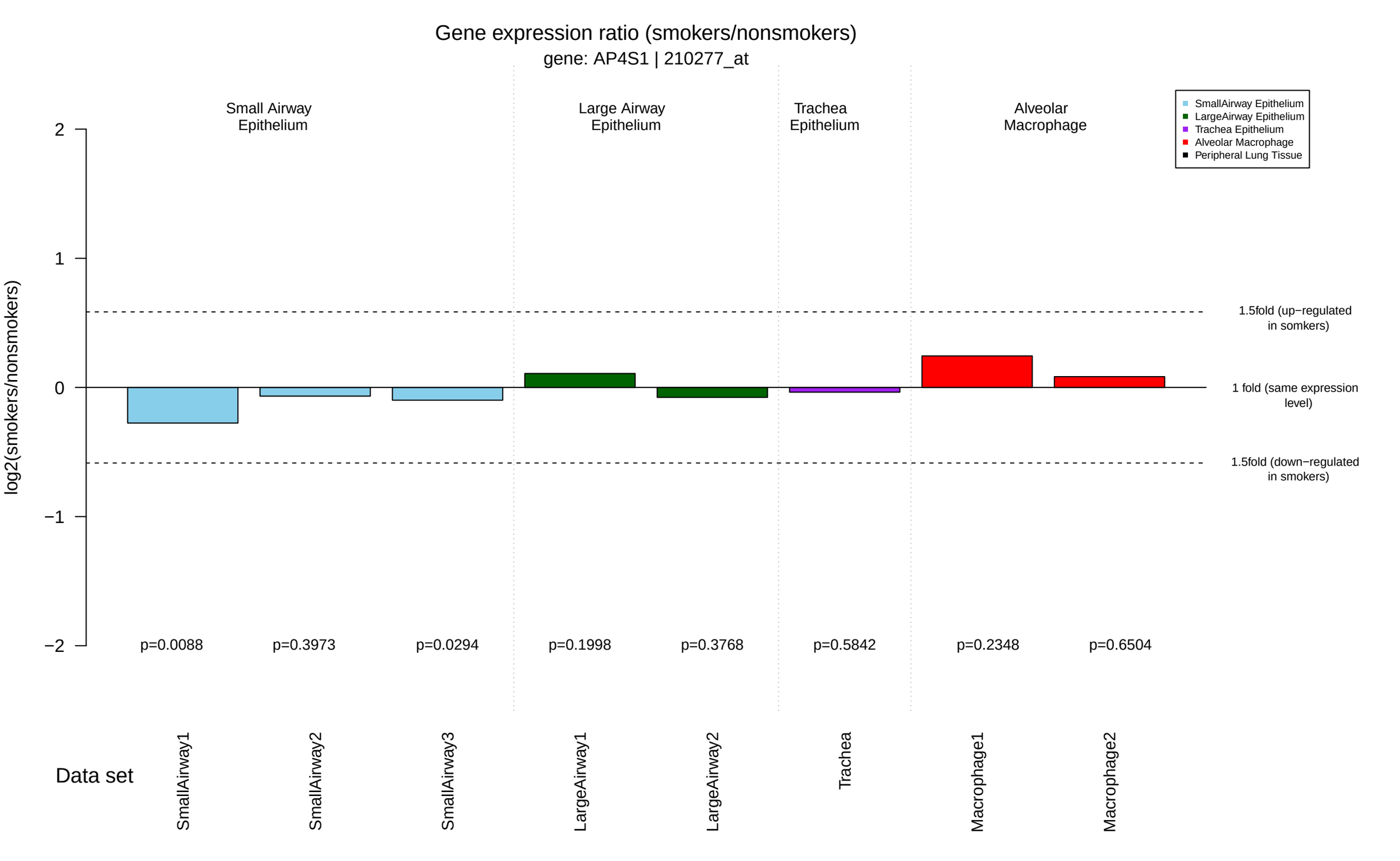
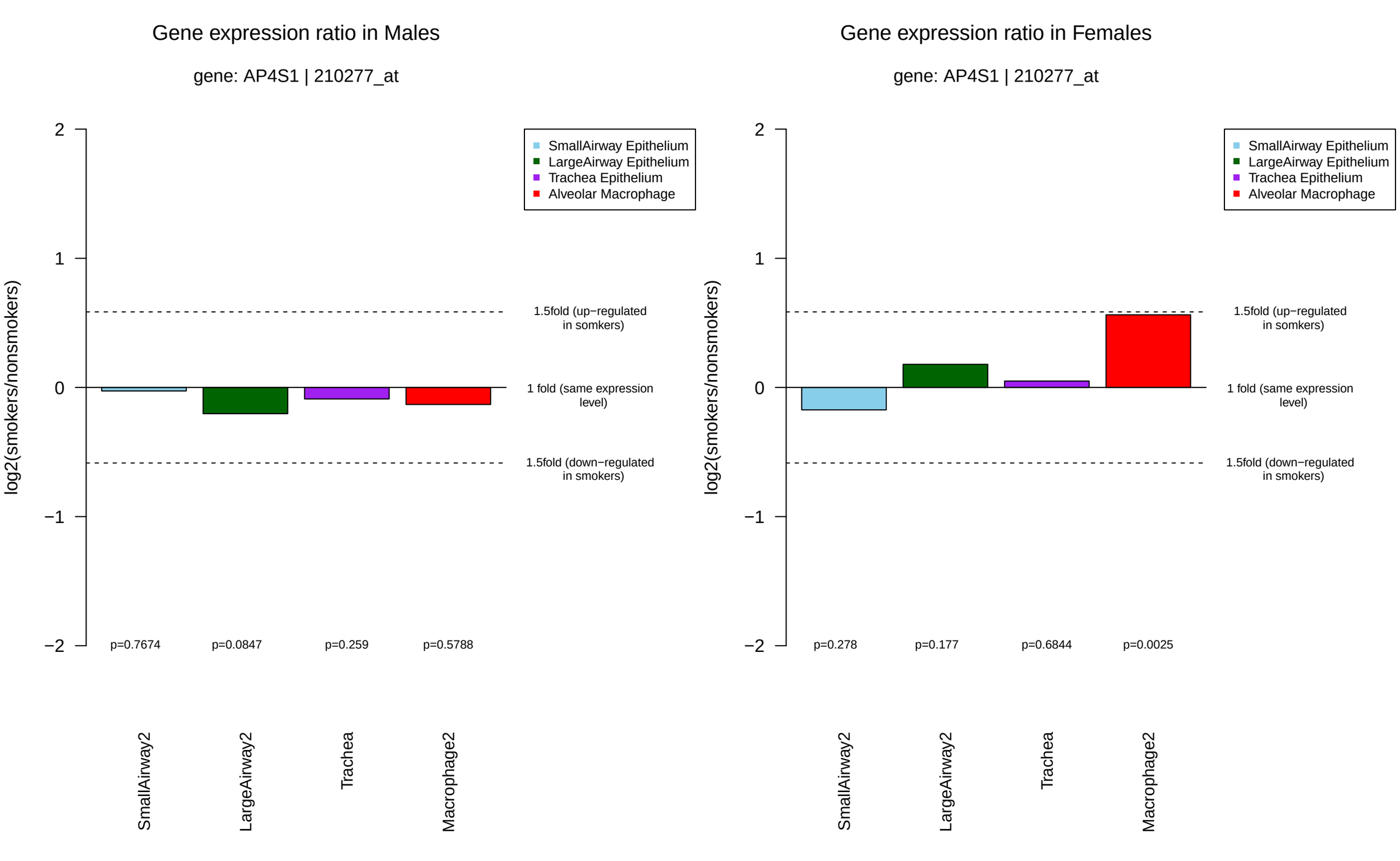
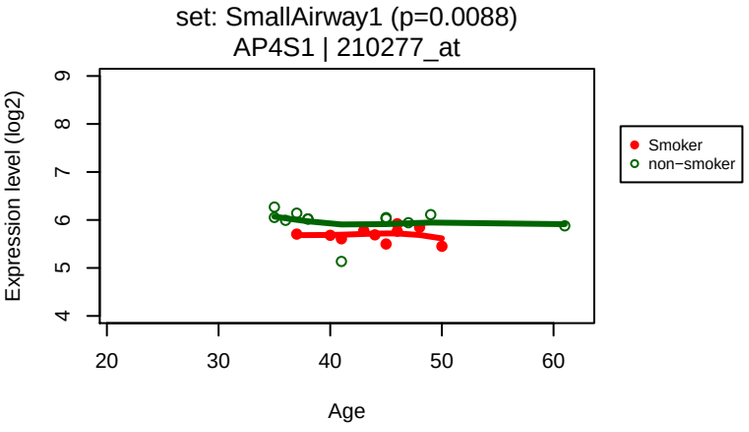
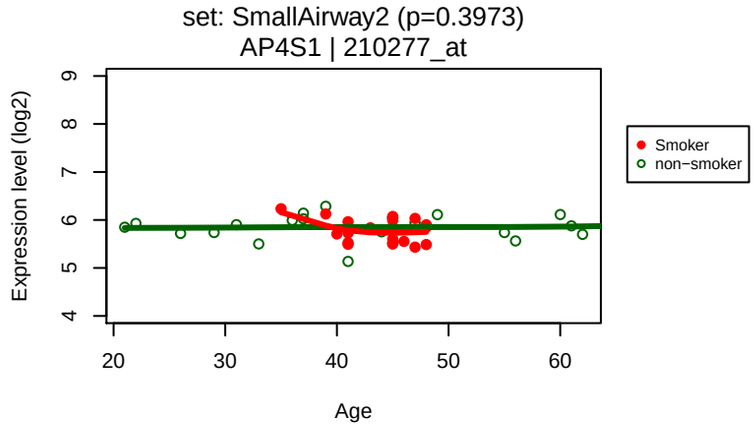
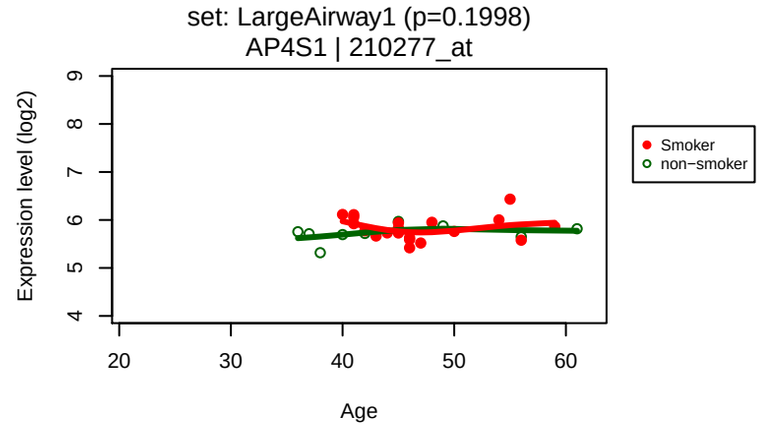

Gene expression of smokers and nonsmokers
gene: AP4S1 | 210277_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0088
p=0.3973
p=0.0294
p=0.1998
p=0.3768
p=0.5842
p=0.2348
p=0.6504
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

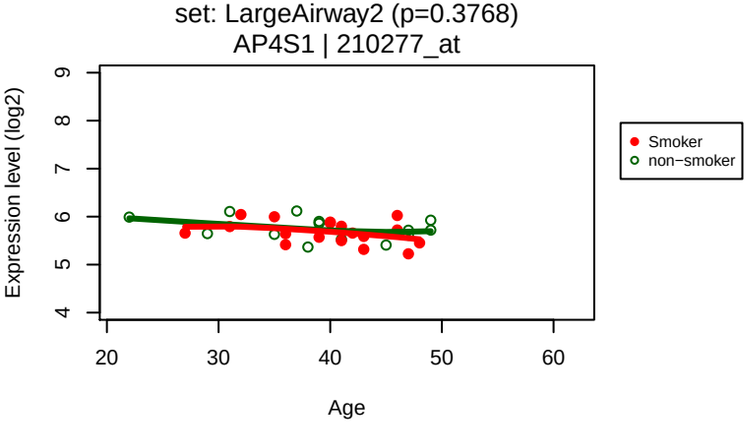

Gene expression ratio (smokers/nonsmokers)
gene: AP4S1 | 210277_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0088
p=0.3973
p=0.0294
p=0.1998
p=0.3768
p=0.5842
p=0.2348
p=0.6504
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

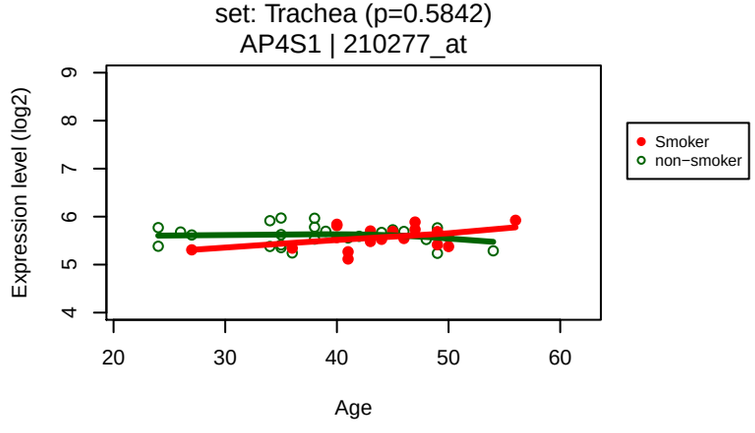

Gene expression ratio in Males
gene: AP4S1 | 210277_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7674
p=0.0847
p=0.259
p=0.5788
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP4S1 | 210277_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.278
p=0.177
p=0.6844
p=0.0025
SmallAirway2
LargeAirway2
Trachea
Macrophage2

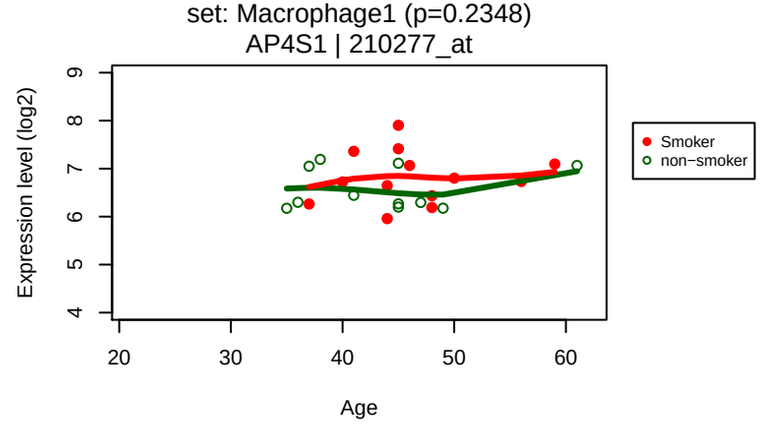

set: SmallAirway1 (p=0.0088)
AP4S1 | 210277_at
Expression level (log2)
Age
Smoker
non−smoker

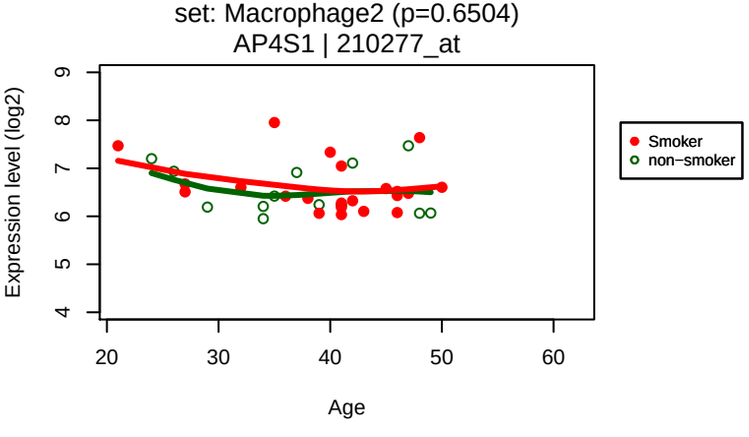

set: SmallAirway2 (p=0.3973)
AP4S1 | 210277_at
Expression level (log2)
Age
Smoker
non−smoker

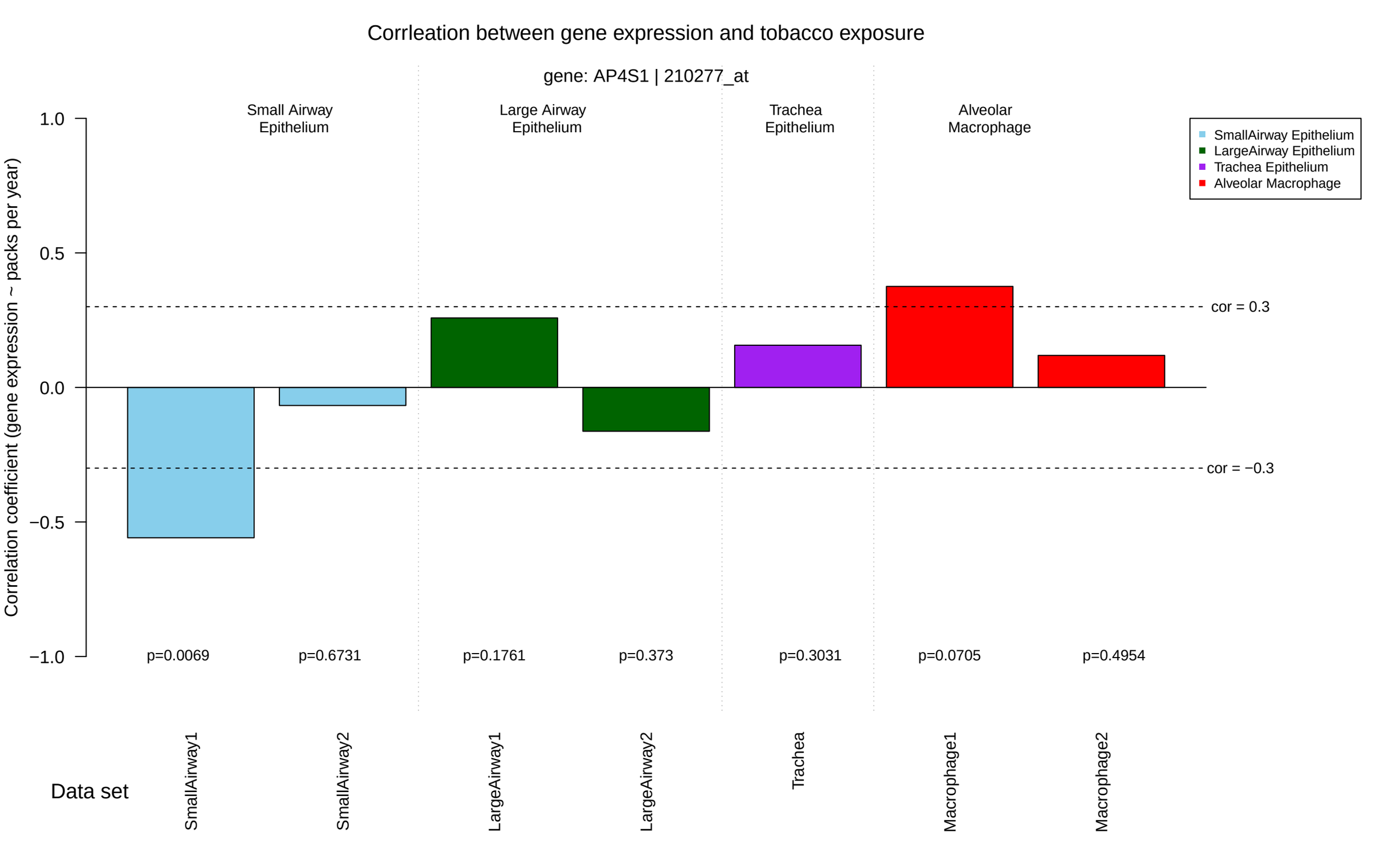

set: LargeAirway1 (p=0.1998)
AP4S1 | 210277_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3768)
AP4S1 | 210277_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5842)
AP4S1 | 210277_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2348)
AP4S1 | 210277_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6504)
AP4S1 | 210277_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP4S1 | 210277_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0069
p=0.6731
p=0.1761
p=0.373
p=0.3031
p=0.0705
p=0.4954
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2